

- 2 Yeasts are unicellular organisms from the kingdom Fungi. *Saccharomyces cerevisiae* is one species of yeast that can carry out either asexual reproduction by mitosis or sexual reproduction by meiosis.

Budding in *S. cerevisiae* is a process where a small daughter cell forms as a bud on the parent cell. The bud contains a copy of the parent cell nucleus and it eventually separates from the parent cell to form a new cell.

S. cerevisiae can exist in two forms: haploid cells or diploid cells.

- Haploid cells can be one of two different mating types: **a** and **α**.
- Haploid cells can only mate with other haploid cells of the opposite mating type.

Fig. 2.1 shows the life cycle of *S. cerevisiae* with its asexual and sexual reproductive stages.

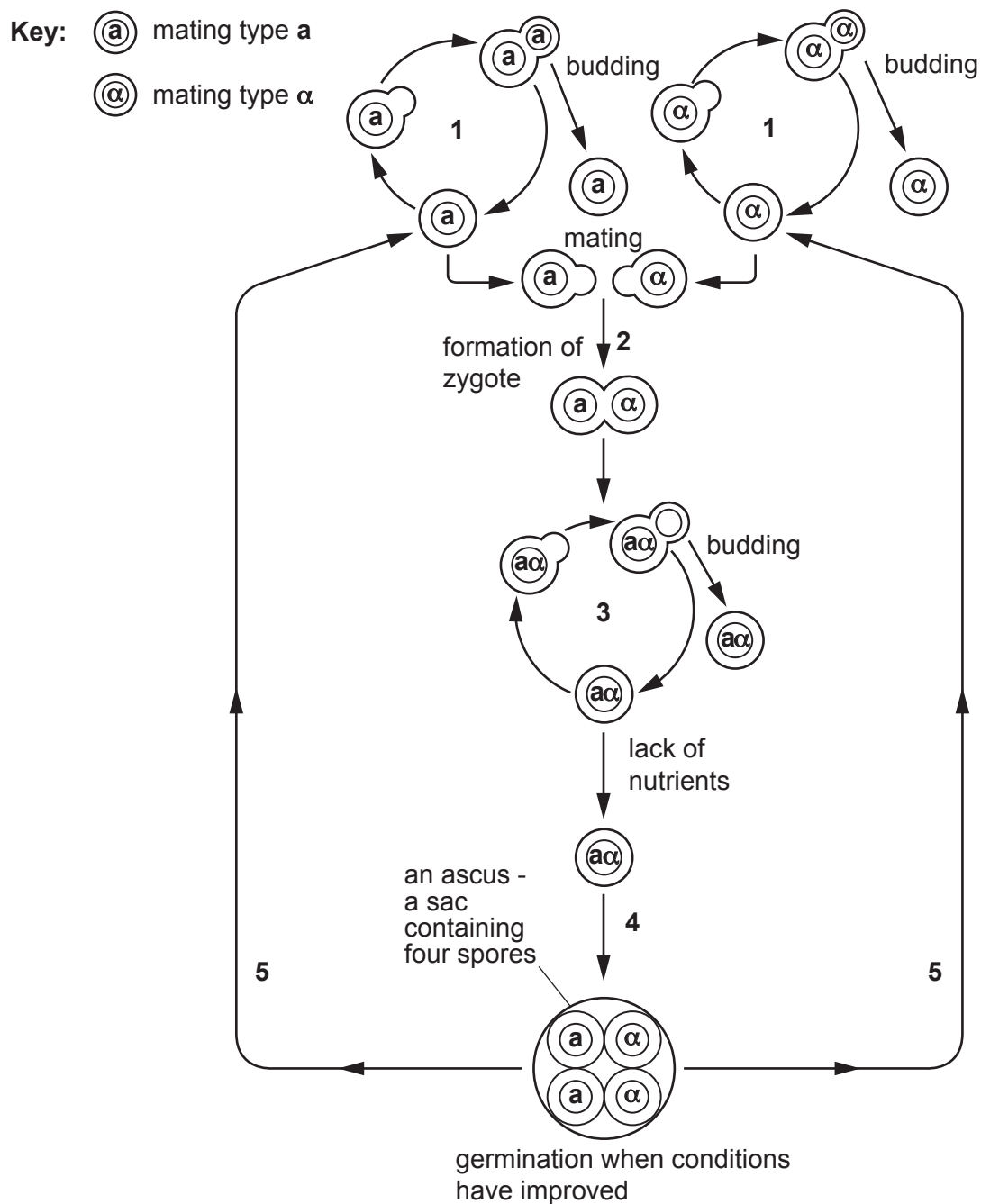


Fig. 2.1

(a) With reference to Fig. 2.1, state the numbers of the stages 1–5 that:

involve mitosis

involve meiosis

produces new genetic variation

shows only haploid cells

shows only diploid cells

[5]

(b) When there is a lack of nutrients, cells made in stage 3 will carry out stage 4 to make spores, which germinate only when conditions improve.

Suggest **and** explain how the type of reproduction that makes spores during stage 4 is advantageous for *S. cerevisiae* in a changing environment.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(c) Haploid and diploid cells of *S. cerevisiae* can carry out **asexual** reproduction.

Suggest why a new harmful recessive mutation may **not** have a damaging effect on:

- an asexually reproducing population of haploid cells of *S. cerevisiae*
- an asexually reproducing population of diploid cells of *S. cerevisiae*.

.....

.....

.....

.....

.....

.....

..... [3]

(d) State **two** features, other than reproduction using spores, of the kingdom Fungi.

.....

.....

..... [2]

[Total: 14]